

Data-Driven Investment Intelligence Using NIFTY-50 Market Data

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1. Introduction

Financial markets generate massive amounts of data every day, making it increasingly difficult for investors to identify meaningful patterns and make informed decisions. While stock market data is publicly available, transforming raw historical prices and trading volumes into actionable insights requires sophisticated analytical techniques.

This project presents an AI-powered investment intelligence platform built on historical NIFTY-50 stock market data. The NIFTY-50 index represents fifty of the largest and most influential companies listed on the National Stock Exchange (NSE) of India, spanning sectors such as Banking, Information Technology, Energy, Consumer Goods, Pharmaceuticals, Financial Services, and Manufacturing.

Unlike traditional stock price prediction competitions, the focus of this project is on building a practical decision-support system that combines machine learning, financial analytics, and explainability to assist investors in making informed, data-driven decisions.

The platform addresses three core capabilities: a Stock Predictor Engine for forecasting future price behavior, a Portfolio Construction Module for generating investor-profile-based allocations, and a Risk Assessment Module for evaluating historical risk at both stock and portfolio levels.

2. Dataset Overview

Primary Dataset: NIFTY-50 Stock Market Dataset **Source:**

<https://www.kaggle.com/datasets/rohanrao/nifty50-stock-market-data/data>

The dataset contains daily market records for 49 companies that have been part of the NIFTY-50 index of the National Stock Exchange of India. One file (NIFTY50_all) was identified as the index-level aggregated file and excluded from individual stock analysis to avoid data contamination.

Dataset Statistics:

Parameter	Value
Total Records	470,384 (raw), 235,192 (after filtering index file)
Stocks Covered	49 individual companies
Date Range	January 2000 – April 2021
Years of Data	~21 years
Records after Feature Engineering	117,840 (after NaN removal)

Columns Available: Date, Symbol, Series, Prev Close, Open, High, Low, Last, Close, VWAP, Volume, Turnover, Trades, Deliverable Volume, %Deliverable

Sector Metadata: Company Name, Industry, Symbol, Series, ISIN Code

The dataset spans 13 distinct industry sectors including Financial Services, Automobile, Consumer Goods, Cement, IT, Pharma, Energy, Metals, Telecom, Construction, Services, Fertilizers, and Media & Entertainment.

3. Exploratory Data Analysis

3.1 Data Quality

Upon loading, the dataset was found to be largely clean. Exactly 50 null values were identified in the Daily Return column, corresponding to one missing value per stock — the first row of each stock's history, which has no previous day to compare against. This is expected and was handled naturally by the feature engineering step. The NIFTY50_all index file was identified and removed before analysis to prevent it from distorting volatility and correlation calculations.

3.2 Return Distribution

The distribution of daily returns across all 49 stocks reveals a classic financial return profile — highly concentrated around zero with rare but extreme outliers in both directions. Most stocks move between -3% and +3% on any given day, with extreme moves beyond

±10% occurring only during major market events such as the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic crash.

3.3 Sectoral Performance

Annualized mean returns by sector reveal significant performance divergence across industries over the 21-year period:

Sector	Annualized Return
Cement & Cement Products	26.09%
Financial Services	26.08%
Automobile	22.36%
Telecom	21.43%
Metals	17.37%
Consumer Goods	17.26%
Fertilizers & Pesticides	14.86%
Construction	13.87%
Services	13.25%
Pharma	10.52%
Energy	10.28%
IT	9.75%
Media & Entertainment	5.45%

Cement and Financial Services were the top-performing sectors at approximately 26% annualized return. Media & Entertainment significantly underperformed at only 5.45%. The IT sector's relatively low ranking (9.75%) despite its reputation for high growth reflects the severe impact of the dot-com crash of 2000–2002, which permanently impaired early IT stock valuations. Energy and Pharma were middle-of-the-pack performers, reflecting commodity price cycles and regulatory pressures, respectively.

3.4 Volatility Analysis

Annualized volatility (standard deviation of daily returns $\times \sqrt{252}$) was computed for each stock:

Most Volatile Stocks:

Stock	Annualized Volatility
VEDL	56.84%
ZEEL	52.58%
JSWSTEEL	52.32%

KOTAKBANK	51.95%
HCLTECH	51.63%

Least Volatile Stocks:

Stock	Annualized Volatility
NESTLEIND	24.23%
POWERGRID	29.71%
COALINDIA	29.83%
NTPC	30.19%
ULTRACEMCO	33.26%

NESTLEIND stands out as a uniquely stable stock — low volatility and strong Sharpe ratio — making it a natural anchor for conservative portfolios. Metals and media stocks (VEDL, ZEEL) showed the highest volatility, reflecting their sensitivity to global commodity cycles and regulatory uncertainty.

3.5 Price Performance and Market Events

Normalized price performance analysis of major stocks (RELIANCE, TCS, HDFCBANK, INFY, WIPRO) indexed to a base of 100 at their respective listing dates reveals that HDFCBANK delivered the most outstanding long-term performance, growing approximately 14 times from its base value. RELIANCE showed strong recovery post-2016. WIPRO and INFY, despite being well-known IT giants, delivered relatively flat performance over the full period due to their heavy losses in the early 2000s.

Two major market crashes are clearly visible in the rolling 30-day average market return chart. The 2008 Global Financial Crisis (September 2008 – March 2009) produced the deepest and most sustained bear period. The COVID-19 crash of February–April 2020 was sharper but shorter, with a rapid V-shaped recovery.

3.6 Correlation Structure

The pairwise return correlation heatmap reveals moderate positive correlations across most NIFTY-50 stocks, which is expected as they all respond to common macroeconomic factors such as interest rates, inflation, and global risk sentiment. Stocks within the same sector (e.g., HDFCBANK and ICICIBANK in Financial Services) show higher correlations than cross-sector pairs. This correlation structure is critical for portfolio construction, as combining low-correlation assets reduces overall portfolio volatility.

4. Feature Engineering

To prepare the data for machine learning models, 15 technical indicators and lag features were engineered for each stock individually, ensuring no data leakage across stocks.

4.1 Trend Indicators

- **MA_20, MA_50:** 20-day and 50-day Simple Moving Averages — capture medium and long-term price trends
- **EMA_12, EMA_26:** Exponential Moving Averages — weight recent prices more heavily than simple averages
- **MACD:** Difference between EMA_12 and EMA_26 — measures momentum shifts
- **MACD_Signal:** 9-day EMA of MACD — generates buy/sell signals when MACD crosses it

4.2 Momentum Indicators

- **RSI (14-day):** Relative Strength Index — measures speed and magnitude of price changes; values below 30 indicate oversold conditions, above 70 indicate overbought

4.3 Volatility Indicators

- **BB_High, BB_Low:** Upper and Lower Bollinger Bands — price envelope based on 2 standard deviations around MA_20
- **ATR:** Average True Range — measures market volatility by decomposing the full range of an asset price

4.4 Volume Indicators

- **OBV:** On-Balance Volume — cumulative volume indicator that relates volume flow to price changes

4.5 Lag Features

- **Return_Lag_1, Return_Lag_3, Return_Lag_5:** Previous 1, 3, and 5 day returns — capture short-term momentum and mean reversion patterns

4.6 Rolling Statistics

- **Volatility_20:** 20-day rolling standard deviation of returns — regime indicator

4.7 Target Variables

- **Next_Return:** Next day's return (regression target)

- **Direction:** Binary indicator — 1 if next day's return is positive, 0 if negative (classification target)

After feature engineering and removal of NaN rows (arising from rolling window calculations), the final dataset contained 117,840 records across 49 stocks with 34 total columns.

5. Stock Predictor Engine

5.1 Methodology

Two complementary prediction tasks were formulated:

1. **Return Prediction (Regression):** Predict the magnitude of next-day return using XGBoost Regressor
2. **Direction Prediction (Classification):** Predict whether the next day's price will go up or down using XGBoost Classifier

XGBoost was selected for its strong performance on tabular financial data, robustness to outliers, built-in regularisation, and interpretability through feature importance scores.

Model Parameters:

- n_estimators: 100
- learning_rate: 0.05
- max_depth: 4
- random_state: 42

Validation Strategy: TimeSeriesSplit with 5 folds was used throughout to prevent data leakage. This ensures the model is always trained on past data and tested on future data, simulating real-world deployment conditions.

5.2 Regression Results

Metric	Value
Mean RMSE (all stocks)	0.02696
Mean MAE (all stocks)	0.01841

Top 10 Most Predictable Stocks (Lowest RMSE):

Stock	MAE	RMSE
HDFCBANK	0.01106	0.01682
POWERGRID	0.01327	0.01726
HINDUNILVR	0.01384	0.01859

ITC	0.01340	0.01923
BAJAJ-AUTO	0.01497	0.01941
HEROMOTOCO	0.01482	0.02005
HDFC	0.01521	0.02018
ULTRACEMCO	0.01506	0.02021
SHREECEM	0.01549	0.02026
NESTLEIND	0.01549	0.02036

HDFCBANK was the most predictable stock with RMSE of 0.01682, likely due to its stable institutional ownership, consistent dividend policy, and lower sensitivity to commodity price swings.

5.3 Classification Results

Metric	Value
Mean Directional Accuracy	50.00%

Top 10 by Directional Accuracy:

Stock	Directional Accuracy
HCLTECH	52.37%
TITAN	52.37%
ICICIBANK	51.49%
AXISBANK	51.34%
POWERGRID	51.34%
SUNPHARMA	51.34%
EICHERMOT	51.00%
JSWSTEEL	50.81%
BAJFINANCE	50.81%
HINDUNILVR	50.71%

A mean directional accuracy of 50% across all stocks is an expected and honest result. Financial markets are near-efficient, meaning that on average, price direction is as difficult to predict as a coin flip. Models claiming 70%+ directional accuracy on out-of-sample data are almost certainly overfitting. The marginal edge of 51–52% for top stocks like HCLTECH and TITAN represents a statistically meaningful signal that can be exploited in a systematic trading strategy.

6. Portfolio Construction Module

6.1 Methodology

Mean-Variance Optimization (MVO), originally proposed by Harry Markowitz, was used to construct portfolios. The optimization uses 5 years of recent data (2016–2021) to estimate expected returns and the covariance matrix, which are then used to find optimal portfolio weights.

Three investor profiles were modelled:

Conservative: Minimize portfolio volatility — suitable for risk-averse investors who prioritize capital preservation over returns.

Balanced: Equal weight across all 49 stocks — no assumptions about future performance, maximum diversification.

Aggressive: Maximize Sharpe Ratio — suitable for growth-oriented investors willing to accept higher volatility for better risk-adjusted returns.

Constraints applied: minimum 1% and maximum 20% allocation per stock to ensure diversification.

Risk-free rate used: 6% per annum (Indian Government Bond yield approximation).

6.2 Portfolio Performance

Portfolio	Annualized Return	Volatility	Sharpe Ratio
Conservative	13.27%	15.84%	0.459
Balanced	12.57%	18.96%	0.346
Aggressive	23.90%	21.37%	0.838

6.3 Portfolio Allocations

Conservative Portfolio — Top Holdings: NESTLEIND (10.8%), POWERGRID (10.4%), HINDUNILVR (8.7%), DRREDDY (8.3%), CIPLA (6.3%), TCS (6.2%), ITC (4.2%)

The conservative portfolio naturally gravitated toward defensive, low-volatility stocks — consumer staples (NESTLEIND, HINDUNILVR), utilities (POWERGRID), and pharmaceuticals (CIPLA, DRREDDY). These sectors maintain stable revenues regardless of economic cycles.

Aggressive Portfolio — Top Holdings: BAJAJFINSV (20%), TITAN (20%), NESTLEIND (7.4%), HINDALCO (5.3%), TATASTEEL (2.9%)

The aggressive portfolio concentrated in high-growth financial services (BAJAJFINSV) and consumer discretionary (TITAN), which delivered the highest Sharpe ratios over the sample period. The Sharpe ratio of 0.838 is considered strong — anything above 0.5 is generally regarded as a good investment on a risk-adjusted basis.

Balanced Portfolio: Equal 2.04% allocation across all 49 stocks — this serves as a benchmark representing the market-cap-weighted index equivalent.

6.4 Key Insight

The aggressive portfolio achieves nearly double the return of the conservative portfolio (23.90% vs 13.27%) at only moderately higher volatility (21.37% vs 15.84%), resulting in a much superior Sharpe ratio (0.838 vs 0.459). This demonstrates that active stock selection, guided by historical risk-return optimisation, can meaningfully outperform naive equal-weight allocation.

7. Risk Assessment Module

7.1 Metrics Computed

For each of the 49 stocks, the following risk metrics were computed using the full historical dataset:

- **Annualized Volatility:** Standard deviation of daily returns scaled to annual frequency ($\times \sqrt{252}$)
- **Sharpe Ratio:** (Annualized Return – Risk-Free Rate) / Annualized Volatility, using 6% risk-free rate
- **Sortino Ratio:** Like Sharpe but penalizes only downside volatility — more appropriate for asymmetric return distributions
- **Maximum Drawdown:** Largest peak-to-trough decline in cumulative returns — represents worst-case loss scenario
- **Value at Risk (95%):** The daily return threshold below which losses fall only 5% of the time

7.2 Top Stocks by Sharpe Ratio

Stock	Volatility	Sharpe	Sortino	Max Drawdown	VaR 95%
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SHREECEM	38.97%	0.827	1.297	-79.97%	-3.54%
BAJFINANCE	48.66%	0.678	0.813	-93.28%	-3.85%
BAJAJFINSV	40.09%	0.656	0.923	-86.79%	-3.49%
MARUTI	34.48%	0.604	0.898	-61.34%	-3.17%
EICHERMOT	48.04%	0.598	0.742	-93.82%	-3.73%
NESTLEIND	24.23%	0.569	0.891	-32.40%	-2.15%
ULTRACEMCO	33.26%	0.568	0.871	-77.78%	-3.06%

7.3 Least Risky Stocks

Stock	Volatility	Sharpe	Max Drawdown
NESTLEIND	24.23%	0.569	-32.40%
POWERGRID	29.71%	0.145	-64.12%
COALINDIA	29.83%	-0.360	-75.07%
NTPC	30.19%	0.015	-73.16%
ULTRACEMCO	33.26%	0.568	-77.78%

7.4 Worst Maximum Drawdowns

Stock	Max Drawdown
VEDL	-98.58%
WIPRO	-98.31%
HINDALCO	-97.46%
INFY	-96.88%
HCLTECH	-96.82%

7.5 Key Risk Insights

NESTLEIND is the standout defensive stock — it has the lowest volatility (24.23%), the smallest maximum drawdown (-32.40%), and yet maintains a healthy Sharpe ratio of 0.569. It is the only stock that combines low risk with strong risk-adjusted returns.

COALINDIA and **WIPRO** both have negative Sharpe ratios, meaning they failed to beat the risk-free rate of 6% on a risk-adjusted basis over the full 21-year period — they destroyed wealth in real terms.

VEDL experienced a near-total drawdown of -98.58% at its worst point, illustrating the extreme risk of concentrated positions in commodity-linked stocks.

The mean Sharpe Ratio across all stocks is 0.27, indicating that on average, NIFTY-50 stocks provide modest but positive risk-adjusted returns above the risk-free rate over the long term.

8. Explainability and Feature Importance

8.1 Methodology

To ensure transparency in the model's decision-making process, XGBoost's built-in feature importance scores were computed for the most predictable stock (HDFCBANK). Feature importance in XGBoost measures how frequently each feature is used to split decision trees, weighted by the improvement in prediction accuracy it provides.

8.2 Results

Feature	Importance Score
BB_Low	62.18%
MACD_Signal	19.41%
EMA_12	2.36%
EMA_26	2.14%
Return_Lag_1	1.62%
ATR	1.55%
Return_Lag_3	1.42%
Volatility_20	1.35%

8.3 Interpretation

BB_Low (62.18%) is by far the most predictive feature. The Lower Bollinger Band represents a price level of approximately 2 standard deviations below the 20-day moving average. When the price approaches or breaches this level, it signals an oversold condition and historically precedes a mean reversion bounce. This finding supports the view that short-term price reversion is a real and exploitable pattern in HDFCBANK's price history.

MACD_Signal (19.41%) is the second most important feature. The MACD Signal line captures medium-term momentum shifts. When MACD crosses above its signal line, it indicates building upward momentum, and vice versa.

EMA_12 and EMA_26 contribute modestly (~2% each), suggesting that the level of the price relative to its moving averages matters less than the rate of change captured by MACD.

RSI, MA_20, MA_50 contribute very little (under 1% each), indicating that simple price level indicators add limited predictive value beyond what is already captured by Bollinger Bands and MACD for this stock.

9. Conclusion

This project successfully developed an AI-powered investment intelligence platform using 21 years of NIFTY-50 historical data. The platform delivers three mandatory modules — stock prediction, portfolio construction, and risk assessment — as well as model explainability.

Key Findings:

The Stock Predictor Engine achieved a mean RMSE of 0.02696 across 49 stocks using XGBoost with time-series cross-validation. HDFCBANK was the most predictable stock (RMSE 0.01682). The mean directional accuracy of 50% is consistent with the efficient market hypothesis and represents an honest evaluation of the model's capabilities.

The Portfolio Construction Module demonstrated that optimized portfolios significantly outperform naive equal-weight allocation. The aggressive portfolio achieved a Sharpe ratio of 0.838 with 23.90% annualized return, compared to 0.346 for the balanced portfolio.

The Risk Assessment Module revealed wide dispersion in risk-adjusted returns across NIFTY-50 stocks. NESTLEIND stands out as the best risk-adjusted investment over the 21-year period, while COALINDIA, WIPRO, and INFY destroyed wealth on a risk-adjusted basis.

Feature importance analysis revealed that Bollinger Band signals (BB_Low) and MACD momentum (MACD_Signal) are the dominant drivers of next-day price prediction for HDFCBANK, accounting for over 80% of model importance.

Limitations:

The models are trained on historical data and assume that past patterns repeat in the future — a strong assumption in financial markets. The dataset ends in April 2021, and post-pandemic structural shifts in the Indian economy may render some historical patterns less applicable. The 6% risk-free rate assumption may not hold across the full 2000–2021 period, during which Indian bond yields varied between 5% and 9%.

Reference : NIFTY-50 Stock Market Dataset. Kaggle, 2021.